SSEE and the Growth Sector Re-examined: No Second Matter Component, and No S_8 Tension, on Raw Survey Data

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

October 2026 (SSEE, Paper 6 of 10)

Abstract

We test the growth-of-structure sector of SSEE against *raw* survey data rather than against compressed statistics whose reduction assumes a ΛCDM background. The model has a single matter density, $\Omega_m = 0.308881$, entering the background and the growth of structure alike; the background is fixed by the algebra. Against the 357 raw $\xi_\pm$ points of KiDS-Legacy [Wright et al., 2025], with the amplitude free, the shear prefers $\log(10^{10}A_s) = 3.0544 \pm 0.0383$, -0.25σ from the 3.04483 that SSEE's own CMB fit supplies (under the identical pipeline with the Planck background, 0.16σ). Fixing the amplitude to the CMB value leaves the growth sector with *no free cosmological parameter* and costs almost nothing: $\chi^2_{\rm min} = 415.59$ against 414.99 with the amplitude free, $\Delta\mathrm{BIC} = 5.28$ in favour of fixing it, and the nuisance parameters do not move to compensate. $S_8 = 0.8273$ is then an output of the algebra and the CMB, against a measured 0.8265 ± 0.0176. KiDS-Legacy predates this analysis by sixteen months and no temporal priority is claimed: the result is a consistency of A_s between epochs. We state against our own interest that the three χ^2 values compared span only 0.74: cosmic shear does not discriminate between these backgrounds, and what separates them is the parameter count. (The earlier KiDS-1000 release is superseded and kept only as the record of how the retraction was reached.) Against the raw BOSS DR12 multipoles (144 points, one-loop LPT), after a ΛCDM control with its background free reproduces the published full-shape analysis of the same data (every parameter within the pre-set 0.3σ), SSEE with every cosmological parameter fixed — amplitude included — gives $\chi^2_{\rm min} = 117.10$ against 114.26 for ΛCDM with three more free parameters ($\Delta\chi^2 = +2.84$, $\Delta\mathrm{BIC} = -12.1$), and 115.34 for ΛCDM fixed at Planck. This paper also retracts a second matter sector and an associated particle ($m_\phi = 40.70$ eV) proposed in an earlier version: the sector was

defined by subtracting $1 + w_0 = 0.160050$ — a number of the dark-energy equation of state, not a density — from the matter density, and the S_8 tension it was built to close existed only in the compressed statistic, never in the raw data.

Contents

1 Introduction: why the growth sector needed re-examination **4**

2 The retraction: why the second sector is withdrawn **4**

2.1 The subtraction that never was a density 4

2.2 Falsifiability is not evidence of existence 5

2.3 Open problems closed by dissolution 5

3 Methodology: measuring against raw data, not compressed statistics **6**

3.1 The principle: combined data reports its own internal tensions 6

3.2 Test design: ingredients transported, not re-fitted 6

3.3 Raw KiDS-1000 cosmic shear likelihood (superseded release) 7

3.4 Raw BOSS DR12 redshift-space multipoles 7

4 S_8 against raw cosmic shear: KiDS-Legacy **8**

5 The earlier KiDS-1000 release (superseded, for the record) **10**

5.1 Historical note: the particle's exclusion, for the record 13

6 $f\sigma_8$: growth rate against raw BOSS DR12 clustering **13**

6.1 Why the exploratory linear-Kaiser stage does not publish 14

7 Conclusions **15**

1 Introduction: why the growth sector needed re-examination

The background sector of SSEE — the algebraic derivation of ω_b, ω_c, n_s, w_0 and w_a from φ and π, with the global H_0 the output of the SH0ES screening cascade (Papers 9–10) — has been tested against Planck 2018, DESI DR2 BAO, and Pantheon+ in Papers 1–4 and 9–10. The growth-of-structure sector — how matter perturbations evolve once the background is fixed — is tested independently, against two late-time probes: weak gravitational lensing (cosmic shear: KiDS-1000 and KiDS-Legacy) and redshift-space galaxy clustering (BOSS DR12). Both probes report their results as derived quantities: KiDS publishes $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$, and BOSS publishes $f\sigma_8(z)$, the growth rate times the amplitude of clustering.

Both derived quantities share a hidden dependency that matters for a model whose background departs from ΛCDM: converting a raw two-point statistic (shear correlation functions, or galaxy power-spectrum multipoles) into a compressed number requires assuming a fiducial cosmology — the Alcock–Paczynski effect [Alcock and Paczynski, 1979] for redshift-space clustering, and the non-linear matter power spectrum template for lensing. Both official pipelines assumed ΛCDM calibrated to Planck [Planck Collaboration, 2020]. Comparing a ΛCDM-conditioned number against a background that is *not* ΛCDM (SSEE has a Chevallier–Polarski–Linder [Chevallier and Polarski, 2001, Linder, 2003] equation of state $w_0 = -0.839950$, not -1) imports part of the ΛCDM background into the comparison itself.

This paper re-does both measurements on the raw data products — the KiDS-Legacy $\xi_\pm(\theta)$ (Wright et al. 2025 [Wright et al., 2025]; the superseded KiDS-1000 release, Asgari et al. 2021 [Asgari et al., 2021], is kept for the record) and BOSS DR12 power-spectrum multipoles $P_0, P_2(k)$ (Beutler & McDonald 2021 [Beutler and McDonald, 2021]) — applying the model-dependent deformation (Alcock–Paczynski for BOSS; the non-linear matter power spectrum template for KiDS) to the *theory prediction*, never to the data. This keeps the measurement clean of any assumed background, at the cost of re-deriving both likelihoods from the covariance matrices up rather than reusing the compressed public numbers.

2 The retraction: why the second sector is withdrawn

2.1 The subtraction that never was a density

The two-sector construction, retired here, rested on

$$\Omega_{\phi\mathrm{DM}} \;=\; \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}}, \qquad \Omega_{m,\mathrm{dyn}} \equiv 1 + w_0. \tag{1}$$

$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ is a measured density fraction, fixed by the CMB-calibrated background. $\Omega_{m,\mathrm{dyn}} = 1 + w_0 = 0.160050$ is, by construction, one plus the dark-energy

equation of state (the saturation s_m of Paper 1) — an algebraic identity of the background sector's $w_0 = -T_R/M_V$, entering the Friedmann equations as an equation-of-state parameter, never as a density. Both quantities are dimensionless, so the subtraction is arithmetically well defined; but a well-defined subtraction between a density and an equation-of-state number does not itself define a density, let alone a particle. The internal code comments that introduced this construction stated explicitly that $\Omega_{m,\mathrm{dyn}}$ "is NOT a matter density: it is the dynamical cold sector $(1 + w_0)$" in the same breath as summing it with $\Omega_{\phi\mathrm{DM}}$ to recover $\Omega_{m,\mathrm{CMB}}$ — an internal inconsistency once both statements are read together. ($1 + w_0$ is dimensionless and is, by construction, one plus the dark-energy equation of state; the parenthetical gloss is ours, not the source comment's.)

2.2 Falsifiability is not evidence of existence

A canonical particle mass, $m_\phi = 40.70$ eV — retracted here — was derived from this subtraction via a fixed algebraic multiplier (retracted: $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS_V} = 594.28$) acting on $\Sigma m_\nu^{\mathrm{active}}$, explicitly to render the two-sector hypothesis falsifiable against free-streaming suppression in the matter power spectrum. Falsifiability is a necessary property of a scientific hypothesis; it is not evidence that the hypothesis's object exists. Once the subtraction that defined $\Omega_{\phi\mathrm{DM}}$ is shown not to correspond to a physical density, the particle built on top of it has nothing left to be made of, and is withdrawn together with it — independently of whatever bound the data would place on its mass. §5 reports that bound only as part of the historical record of how the retraction was reached, not as a surviving result.

2.3 Open problems closed by dissolution

Four open problems catalogued against the retired two-sector construction close by dissolution, following the precedent of OP-8 (the analogous Ω_m partition via the algebraic constant MIRA, retired 2026-06-18):

- **OP-9** (origin of the $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$ coefficient) — closes: there is no coefficient left to derive.
- **OP-10** (the "single free scalar field" claim, admitted as "technically violated" by the retired two-sector construction) — closes, and the single-field philosophy of SSEE is restored without qualification.
- **OP-11** (ξ, the non-minimal coupling of the second sector — an explicit additional free continuous parameter) — closes: the parameter is removed, not fitted.
- **OP-12** (T_ϕ, described in its own documentation as "bookkeeping, not a temperature derived from physics") — closes.

No new open problem is created by this retraction: what remains is $\Omega_m = 0.308881$, a single measured quantity, entering both the background and the growth-of-structure sector with no partition.

3 Methodology: measuring against raw data, not compressed statistics

3.1 The principle: combined data reports its own internal tensions

A dataset built by combining independent probes under a flexible model does not report fidelity to any one of them: it reports the point at which their mutual tensions are minimized, which is not the same thing. A model with enough freedom to fit each probe separately with a *different* set of nuisance values will inevitably report a lower χ^2 when its parameters are free to migrate between probes than when they are held fixed to what any single probe prefers —not because the combination is more accurate, but because it is averaging disagreement rather than resolving it. This is the generic mechanism behind the well-documented S_8 tension between Planck and cosmic shear surveys under ΛCDM [Di Valentino et al., 2021]: parameters fitted to the CMB [Planck Collaboration, 2020], carried over unchanged to lensing, land outside the region preferred by KiDS-1000 [Asgari et al., 2021, Heymans et al., 2021] and DES Y3 [Amon et al., 2022] alike.

SSEE is structurally unable to exploit this mechanism: its background $(\omega_b, \omega_c, h, n_s, w_0, w_a)$ is fixed by algebra, identical whether evaluated against the CMB, cosmic shear, or redshift-space clustering. The only cosmological parameter that could vary between probes is the primordial amplitude A_s (equivalently $\log A$). In SSEE it is counted once, at the CMB, and carried to every other probe fixed (§4).

3.2 Test design: ingredients transported, not re-fitted

To make this principle falsifiable rather than rhetorical, we perform the following test (a design due to M. Almeida): the three probes — Planck CMB, KiDS-Legacy shear, DESI DR2 BAO — are combined with the shared amplitude free, yielding the ingredient set the *combination* prefers. That same ingredient set is then evaluated against each probe *individually*, without re-fitting. Transporting parameters from a joint fit to an individual dataset always degrades χ^2 relative to a dedicated fit — that is arithmetic, true of any model, and proves nothing by itself. The diagnostic quantity is how much it degrades relative to what the probe's own statistical uncertainty predicts:

$$\Delta\chi^2(\text{probe}) \;=\; \chi^2\Big(\text{joint ingredients}\Big) - \chi^2\Big(\text{probe-fitted ingredients}\Big). \tag{2}$$

For SSEE, $\Delta\chi^2 \equiv 0$ by construction on every background parameter: there is no ingredient to transport, because none of $\omega_b, \omega_c, h, n_s, w_0, w_a$ are free to begin with. The one transported quantity is A_s, taken from the CMB; for the CMB that is its own fit, BAO does not depend on it, and for the shear the cost is measured (§4). For ΛCDM, $\Delta\chi^2 > 0$ is measured, not assumed. Results in Table 1.

Table 1: Ingredient transport test (§3.2). $\Delta\chi^2$ evaluated per probe, joint-fit ingredients vs. probe-dedicated fit. ΛCDM: total loss 9.08 for 8 degrees of freedom (0.96σ), from the joint CMB+KiDS-Legacy+DESI fit (`lcdm_conjunta.json`).

	CMB (Planck)	shear (KiDS-Legacy)	BAO (DESI DR2)
ΛCDM, $\Delta\chi^2$	3.18	3.11	2.79
SSEE, $\Delta\chi^2$	0 (own fit)	+0.601 (fixing A_s)	0 (no A_s)

3.3 Raw KiDS-1000 cosmic shear likelihood (superseded release)

The current S_8 result is on KiDS-Legacy, whose likelihood settings and control are given in §4. This subsection describes the setup of the earlier KiDS-1000 analysis, kept for the record (§5).

We evaluate χ^2 against the 225 (of 270) scale-cut $\xi_\pm(\theta)$ data points from the KiDS-1000 Blind C release (Asgari et al. 2021 [Asgari et al., 2021]), using the flat-sky Limber approximation for the shear power spectrum C_ℓ^{ij} (GG+GI+II, NLA intrinsic alignment model, Bridle & King 2007 [Bridle and King, 2007]), HMcode-2015 non-linear matter power spectrum (Mead et al. 2015 [Mead et al., 2015]) with the one-parameter baryon feedback prescription used in the official KiDS-1000 analysis, and the official redshift-distribution SOM covariance and δ_c priors. The pipeline reproduces the published maximum-posterior likelihood to 0.3% (control: $\chi^2 = 261.21$ vs. 260.31 official, the latter from the public nested-sampling chain, MultiNest [Feroz et al., 2009]).

Eight nuisance parameters (baryon feedback amplitude, intrinsic alignment amplitude, five source-redshift shifts, additive c-term) are marginalized via MCMC (Cobaya [Torrado and Lewis, 2021], Metropolis sampler, 4 independent chains, Gelman–Rubin $R-1<0.03$), not minimized, for both SSEE (background fixed, $\log A$ free: 9 parameters total) and ΛCDM (background free: $\omega_b, \omega_c, h, n_s$; 13 parameters total, same nuisance treatment).

3.4 Raw BOSS DR12 redshift-space multipoles

We evaluate χ^2 against the pre-reconstruction power-spectrum monopole and quadrupole $P_0, P_2(k)$ of BOSS DR12 in the two non-overlapping redshift bins ($z_{\rm eff} = 0.38$ and 0.61) $\times$ 2 hemispheres (NGC/SGC), at $0.02 < k < 0.20\,h\,\mathrm{Mpc}^{-1}$: 144 points. The measurements are the re-normalised ones of Beutler & McDonald (2021) [Beutler and McDonald, 2021], which correct a $\sim 10\%$ mismatch between the normalisation of the earlier public spectra [Beutler et al., 2017] and their window function. The window is applied to the model as published with those data: a wide-angle matrix followed by the window matrix, $P^{\rm conv} = W\,M\,P^{\rm model}$. The covariance is the joint NGC+SGC matrix built from the PATCHY mocks by Chen, Vlah & White (2022) [Chen et al., 2022], used as they use it. The Alcock–Paczynski deformation is applied to the *model*, mapped onto the fiducial cosmology ($\Omega_m = 0.31$, $w = -1$) in which the spectra were measured, so the data vector is never touched. Redshift-space distortions are modelled at one loop in Lagrangian perturbation theory, in its effective-field-theory implementation (velocileptors; Baumann et al., 2012, Chen et al.,

2020), on the cold-dark-matter-plus-baryon spectrum. Each (z-bin, hemisphere) carries its own galaxy nuisance parameters — $(1+b_1)\sigma_8$, b_2, b_s sampled, and the counterterms α_0, α_2 and shot noise $\mathrm{SN}_0, \mathrm{SN}_2$, which enter linearly — with the priors of Chen et al. [2022]. This is their setup in every element, chosen so that the pipeline can first be checked against their published result (§6). An earlier linear-Kaiser pass is described in §6.1 and does not publish.

4 S_8 against raw cosmic shear: KiDS-Legacy

The S_8 measurement of this paper is made on the KiDS-Legacy release. The earlier KiDS-1000 release, on which the retraction was first reached, is superseded and kept only as the record in §5. The Planck–shear disagreement it showed was resolved by the KiDS collaboration itself, and not in the model's favour or against it: in the data. KiDS-Legacy [Wright et al., 2025] recalibrates the source redshift distributions and re-analyses the complete survey, and its Appendix I re-runs the KiDS-1000 analysis end to end with the new methods under blinding. The tension against Planck dissolves in steps, quoted there as signed consistencies in S_8 (negative meaning KiDS below Planck): $-2.39\sigma \to -1.42\sigma$ from the $n(z)$ calibration alone, then -1.12σ, -0.85σ and -0.74σ as tomography, alignment model and sky area are updated. Their stated threshold is "the typical KiDS threshold for consistency of 2.3σ (equivalent to a one-sided probability to exceed of 0.01)"; every step with the new calibration falls below it.

Chronology, stated plainly. KiDS-Legacy appeared on arXiv on 2025 March 25, sixteen months before this analysis was written. **We claim no temporal priority over it.** What we report here is an internal consistency test of this model, run on the raw Legacy data vector, and the reader should weigh it as such.

We measure on the 357 unmasked $\xi_\pm$ points of the Legacy release, in three configurations that differ only in what is held fixed. All three use the same likelihood, the same scale cuts (ξ_+: $2'$–$300'$, ξ_-: $4'$–$300'$) and the same eight nuisance parameters.

The control comes first. Before any SSEE number, the shear evaluator is checked against the collaboration's own result, in two steps that cost seconds and hours rather than days. (a) At the maximum-likelihood point of the official chain, with the official intrinsic-alignment model (NLA amplitude per redshift bin), the evaluator returns $\chi^2 = 407.67$ against the published 407.65. (b) Re-weighting a sub-sample of the official posterior (501 effective samples) by the ratio of our likelihood to theirs moves the mean S_8 from 0.8265 to 0.8306, $+0.26\sigma$, inside the 0.3σ criterion fixed before the test. This control exposed two defects in an earlier version of the evaluator — model correlation functions taken at the geometric bin centre instead of the pair-weighted bin average of the release, and a Hankel integral not converged at the largest separations — which together shifted χ^2 at that point by 2.6; the three configurations below were run with the corrected evaluator. On

KiDS-1000 the same correction moves the official-point χ^2 by only 0.05, so the superseded KiDS-1000 chains of §5, run before it, are kept as the record they are and not re-run.

Configuration	background	A_s	free	$\chi^2_{\rm min}$	BIC
SSEE unified	algebraic	from CMB	8	415.59	462.62
SSEE	algebraic	free	9	414.99	467.89
ΛCDM-fixed	Planck 2018	free	9	415.74	468.63

Three results follow, and they answer different questions.

(i) The amplitude the shear asks for is the one the CMB fixes. With the background held by the algebra and $\log A$ free, the Legacy data prefer $\log(10^{10}A_s) = 3.0544 \pm 0.0383$, against the value SSEE's own CMB fit supplies, 3.04483: a separation of -0.25σ. Under the identical pipeline with the Planck background held fixed instead, the same data prefer 3.0378 ± 0.0379 against that background's own CMB value, 0.16σ away. The difference has a mechanism rather than being noise: SSEE's Ω_m is 2.00% lower than Planck's, and the shear asks it for an A_s 1.68% higher — less matter requires more amplitude to produce the same observed clustering. Each background *forces* a different A_s, and the one the algebraic background forces is the one its own CMB predicts. Both $\log A$ values come from the same 357 points and are therefore correlated; no clean significance can be quoted for their difference, and 0.16σ is not a tension either. What can be stated is the direction, and that it has an identified physical cause.

(ii) Fixing A_s costs almost nothing, and could have cost a great deal. Removing the only cosmological parameter from a fit removes its ability to accommodate: if either the algebraic background or the CMB amplitude were wrong, χ^2 rises with nowhere to hide it. It rose by less than one unit. With one parameter fewer, $\chi^2_{\rm min} = 415.59$ against 414.99, and $\Delta{\rm BIC} = 5.28$ in favour of the unified configuration (6.02 against ΛCDM-fixed). A second, independent indication points the same way: the nuisance parameters do not move to compensate. Had the fixed A_s been wrong, $\log T_{\rm AGN}$ and $A_{\rm scale}$ would have run to cover the gap; they shift by 0.059 and 0.04, well inside their own posterior widths (0.21 and 0.72). Removing the parameter raises the minimum by $\Delta\chi^2 = +0.601$, less than one unit: the data do not need A_s as a free parameter. The honest statement is that the cost is small, not that fixing improves the fit.

(iii) S_8 becomes a prediction. With both background and amplitude supplied, $S_8 = 0.8273$ is an output of the algebra and the CMB, with no free cosmological parameter behind it and therefore no posterior width of its own. The official Legacy $\xi_\pm$ posterior, computed from its own published chain as $\sigma_8\sqrt{\Omega_m/0.3}$, gives 0.8265 ± 0.0176. Two caveats bound what this establishes: the agreement in S_8 and the -0.25σ in $\log A$ are the same fact in two variables and do not compound, and the ± 0.0176 width means any prediction within one such width of the measured value would have agreed. The test is not sharp. What it does establish is that the growth sector takes its amplitude from the CMB and describes cosmic shear without adjusting anything.

What this closes. The gap that the retracted light particle was introduced to fill is -0.25σ wide. There is no work left for it to do, and it is withdrawn as unnecessary — not excluded by a bound. Two things follow that are worth stating against our own interest. First, a particle that is not needed may still exist; this is not a constraint on its mass. Second, the three χ^2 values above span only 0.74, so cosmic shear does not discriminate between these backgrounds: what separates them is the parameter count (BIC) and the cross-epoch consistency of A_s, neither of which is a goodness-of-fit statement. **The result is not that SSEE describes cosmic shear better; it is that SSEE describes it equally well without being allowed to fit it.**

5 The earlier KiDS-1000 release (superseded, for the record)

This section records the measurement on the earlier KiDS-1000 release, on which the retraction of §2 was first reached. It is superseded by KiDS-Legacy (§4); nothing below is re-run, and no current result of this paper rests on it.

The SSEE chain (R3) samples nine free parameters — the primordial amplitude $\log A$ plus the eight KiDS-1000 nuisance parameters (the baryonic feedback amplitude $A_{\rm baryon}$, the intrinsic-alignment amplitude $A_{\rm IA}$, five photometric-redshift shifts δz_i, and the additive shear term δ_c) — against the 225 unmasked $\xi_\pm$ data points, with the background held fixed by the algebra ($\Omega_m = 0.308881$, ω_b, ω_c, h, n_s, w_0, w_a all determined, none of them fitted). Four independent MPI chains were run to a Gelman–Rubin convergence of $R - 1 = 0.019$, well inside the stopping criterion $R - 1 < 0.03$, giving 42711 effective samples after discarding the first 30% of each chain. The marginalized posterior is

$$\sigma_8 = 0.7449 \pm 0.0186, \qquad S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3} = 0.7559 \pm 0.0189, \tag{3}$$

with a best-fit $\chi^2 = 265.44$ over 225 data points and 9 free parameters (216 degrees of freedom, $\chi^2/\mathrm{dof} = 1.229$). Both nuisance parameters that carry astrophysical meaning land in unremarkable places ($A_{\rm baryon} = 2.57 \pm 0.31$, $A_{\rm IA} = 0.54 \pm 0.29$), so the fit is not being rescued by a nuisance parameter running to the edge of its prior.

Where the model is actually tested: the fit to the raw data. S_8 is an output of a fit, not a measurement: quoting one model's S_8 against another's compares two *products* of two analyses. The quantity that confronts SSEE with something the model did not produce is the likelihood of the 225 raw $\xi_\pm$ points, so that is where we put the comparison (Table 2). Against the ΛCDM control run through the same pipeline, same mask, same covariance and same nuisance treatment,

$$\Delta\chi^2 \;\equiv\; \chi^2_{\rm SSEE} - \chi^2_{\Lambda\rm CDM} \;=\; 265.44 - 262.75 \;=\; 2.69, \qquad \Delta k = 4, \tag{4}$$

where Δk counts the parameters ΛCDM has and SSEE does not. Four additional free parameters that carried no information would still lower χ^2 by $\langle\Delta\chi^2\rangle = \Delta k = 4$ on average; here they lower it by 2.69, and $p(\Delta\chi^2 \geq 2.69\,|\,\Delta k = 4) = 0.61$. The extra background freedom of ΛCDM is therefore not merely unnecessary on this dataset — it is statistically indistinguishable from absent. Penalising it explicitly gives $\Delta\mathrm{AIC} = -5.31$ and $\Delta\mathrm{BIC} = -19.0$, both favouring SSEE, the latter *very strong* on the scale of Kass and Raftery [1995].

The derived amplitudes are then a consequence rather than the test: $\Delta S_8 = 0.0012$, under a tenth of either error bar. Against the *published* KiDS-1000 value $S_8 = 0.759 \pm 0.024$ the offset is 0.10σ, but that published number is itself ΛCDM-derived, so this second comparison is informative about the pipeline rather than about the cosmology, and we do not rest any claim on it.

The one tension we do report, and it is shared. Both fits are formally poor in absolute terms: $\chi^2 = 265.44$ against an expectation of 216 ± 20.8 for SSEE and 262.75 against 212 ± 20.6 for ΛCDM, i.e. 2.4σ and 2.5σ high, with probabilities to exceed of 0.012 and 0.010. This is not a feature of either cosmology: the KiDS-1000 published best fit on these same points has $\chi^2 = 260.31$, which under any of the plausible effective-parameter counts gives a comparable $p \sim 0.01$–0.03. The excess is a property of the $\xi_\pm$ data vector — plausibly residual small-scale modelling or covariance calibration — and it is inherited identically by every model fitted to it. We flag it rather than absorb it, and note that our control's reproduction of the official value (262.75 vs. 260.31, 0.9%) is what demonstrates the excess is not introduced by our implementation.

What this does and does not claim. It would be wrong to read this as SSEE having carried a 3.5σ tension and then removing it. SSEE never carried that tension. At this level the model has exactly *one* free quantity, the primordial amplitude A_s; the background is fixed by the algebra and is identical in every fit reported here. What varies between determinations is only *which dataset is asked to fix* A_s, and the datasets do not agree with each other:

A_s determined by	$\log(10^{10}A_s)$	S_8	vs. published KiDS-1000[†]
raw KiDS-1000 $\xi_\pm$ (this work)	2.864 ± 0.050	0.7559 ± 0.0189	0.10σ
SSEE fit to raw Planck (full `plik`, B1 chains)	3.042 ± 0.013	0.8261 ± 0.0054	2.73σ
ΛCDM-fitted Planck value (borrowed)	3.045	0.8229 ± 0.0059	2.59σ

[†]The last column compares each row against the *published* KiDS-1000 value, which is a ΛCDM-derived number; it is given for orientation only and no claim here rests on it. The quantity that matters for the argument below is different: the internal disagreement between rows 1 and 2, which are the same model with the same background. In $\log(10^{10}A_s)$ that separation is $3.042 - 2.864 = 0.179 \pm 0.052$, i.e. $\mathbf{3.5}\sigma$.

The first two rows are the same model with the same background, differing only in the data used to set one number, and they sit 3.5σ apart. That separation is a disagreement

between the datasets, not a property of SSEE: it persists when A_s is fixed by SSEE's *own* CMB fit rather than borrowed from ΛCDM, and it is the same Planck–shear discrepancy that ΛCDM also carries. **We do not claim to resolve it.**

What the earlier version of this paper did wrong was to attribute that data–data disagreement to the model — reporting it as "SSEE's S_8 tension" — and then to build a second matter sector and a particle to close it. The disagreement was never the model's to close, and the KiDS collaboration's own recalibration dissolved it in the data (§4).

Table 2: Fit to the *raw* KiDS-1000 $\xi_\pm$ data vector (225 points), primordial amplitude free in both models, MCMC-marginalized. The test is the left block: the likelihood of data neither model produced. The S_8 block is derived output, shown for continuity with the literature. p is the probability to exceed of $\chi^2_{\rm min}$ at the stated dof.

	k	dof	$\chi^2_{\rm min}$	χ^2/dof	p	σ_8	S_8
ΛCDM (background free)	13	212	262.75	1.239	0.010	0.875 ± 0.089	0.7571 ± 0.0194
SSEE (background fixed)	9	216	265.44	1.229	0.012	0.7449 ± 0.0186	0.7559 ± 0.0189
$\Delta\chi^2 = 2.69$ for $\Delta k = 4$ (expected 4 if the extra parameters carry no information, $p = 0.61$); ΔAIC $= -5.31$, ΔBIC $= -19.0$, both favouring SSEE.							
KiDS-1000 published	—	—	260.31	—	—	—	0.759 ± 0.024

The ΛCDM row is the methodological control (R4): the same pipeline, the same raw data, the same nuisance treatment, with the four background parameters $\omega_b, \omega_c, h, n_s$ additionally free. Its purpose is not to provide a new measurement — KiDS-1000 has already published theirs — but to demonstrate that this pipeline reproduces the published ΛCDM result, which is what licenses the SSEE row above it. It does: $S_8 = 0.7571 \pm 0.0194$ against the published 0.759 ± 0.024. We stress that this agreement is *not* a physical measurement of consistency. The published KiDS-1000 value is itself a ΛCDM number — it is what one obtains by fitting ΛCDM to these data — so a ΛCDM run reproducing it tests the code path, not the cosmology. Reading it as a "tension" would be a category error.

What the control licenses is the comparison in Table 2 between the two rows above the rule, which is like for like: identical data, identical code, identical nuisance treatment. The comparison that carries the weight is the χ^2 block, Eq. (4): SSEE pays 2.69 in χ^2 for giving up four parameters, where four uninformative parameters would have been worth 4. The $\Delta S_8 = 0.0012$ in the right-hand block is a restatement of the same fact in derived quantities — under a tenth of either error bar, and, since both fits are to the *same* data, with strongly correlated uncertainties that make the true discrepancy smaller still. SSEE therefore describes the raw shear data as well as ΛCDM does, using four fewer free parameters and with its background fixed algebraically rather than fitted.

Two entries in the ΛCDM row must not be quoted in isolation: $\sigma_8 = 0.875 \pm 0.089$ and $\Omega_m = 0.231 \pm 0.046$ (and correspondingly the broad $\log A$) are individually poorly determined, because cosmic shear constrains essentially only their combination S_8 — which is why that combination is the reported statistic. The published KiDS-1000 analysis shows the same degeneracy.

5.1 Historical note: the particle's exclusion, for the record

Before the subtraction defining $\Omega_{\phi\mathrm{DM}}$ was recognized as physically empty (§2), the two-sector hypothesis was tested directly against raw KiDS-1000 data with the now-retracted particle mass $m_\phi = 40.70$ eV. Free-streaming suppresses σ_8 by 10.4% at that mass, against a raw-data uncertainty of 3.16%; the amplitude A_s cannot compensate because free-streaming changes the *shape* of $P(k)$, not its overall scale. A measured suppression law,$\Delta\sigma_8(\%) = 52088 \cdot m_\phi^{-2.283}$, places a lower bound $m_\phi > 70.3$ eV for consistency with the shear data — 1.73× the canonical mass, and outside the maximal mass reachable by the model's own algebraic lineage grammar (§2, maximum multiplier 594.28). We record this bound as part of how the retraction was reached (§2), not as a live result: the particle it would constrain does not exist in the current model.

6 $f\sigma_8$: growth rate against raw BOSS DR12 clustering

The clustering counterpart of §4 asks the same question of the other growth probe, and the answer is the χ^2 over the 144 raw multipole points, *not* $f\sigma_8$: $f\sigma_8$ is an output of each model, the lesson S_8 taught in §4.

The control comes first. Before any SSEE number, ΛCDM with its background free (Ω_m, H_0, $\ln(10^{10}A_s)$ sampled; ω_b, n_s and $\sum m_\nu$ fixed as in the published analysis) is run through the pipeline of §3.4 and compared with the result Chen et al. [2022] publish for the same multipoles. The criterion, fixed before the run, was every parameter within 0.3 of the published σ (Table 3).

Table 3: Pipeline control: ΛCDM with its background free against the published full-shape analysis of the same BOSS DR12 multipoles. Four MPI chains, $R - 1 = 0.028$.

	Chen et al. [2022]	this pipeline	distance
$\ln(10^{10}A_s)$	2.84 ± 0.13	2.823 ± 0.122	-0.13σ
Ω_m	0.305 ± 0.010	0.304 ± 0.010	-0.10σ
H_0	68.5 ± 1.1	68.38 ± 1.11	-0.11σ
σ_8	0.738 ± 0.048	0.726 ± 0.045	-0.25σ

Getting there corrected three errors in earlier versions of this section, all of which bore on both models alike. (i) The middle BOSS redshift bin overlaps the other two and its public covariance carries no cross terms; using all three counted the shared galaxies twice. (ii) The amplitude was applied as a linear factor on the one-loop output, which is not exact at one loop. (iii) The earlier public spectra carry the $\sim 10\%$ normalisation mismatch corrected by Beutler and McDonald [2021] (we measure $P_0^{\mathrm{old}}/P_0^{\mathrm{new}} = 0.90$–$0.92$), which biases σ_8 low. With the earlier setup the control sat up to 1.1σ from the published values. The $\Delta\chi^2 = -2.03$ in favour of SSEE of an earlier version, and every BOSS number derived from those runs, are withdrawn.

The comparison. With the pipeline validated, SSEE is run with *every* cosmological parameter fixed: the algebraic background and the amplitude its own CMB fit supplies, $\ln(10^{10}A_s) = 3.04483$, exactly as for KiDS-Legacy in §4. Only the galaxy nuisance parameters are fitted. It is compared with the published model, ΛCDM with its background free, and, as a comparison of backgrounds rather than a control, with ΛCDM fixed at Planck 2018 [Planck Collaboration, 2020] ($\ln(10^{10}A_s) = 3.044$). For each, $\chi^2_{\rm min}$ is the raw χ^2 of the 144 points at the best fit (plus the Gaussian prior of the linearly entering nuisance parameters), found from the best chain sample with three different minimisers that agree.

Table 4: BOSS DR12 raw multipoles: fit quality of each model run as itself. k counts every fitted parameter (12 sampled and 16 linear nuisance parameters in all rows, plus the cosmological ones). Δ is relative to ΛCDM with its background free.

	cosmological free	$\chi^2_{\rm min}$	χ^2/dof	$\Delta\chi^2$	ΔBIC
ΛCDM, background free	3	114.26	1.011	—	—
SSEE, all fixed	0	117.10	1.010	+2.84	−12.1
ΛCDM, Planck fixed	0	115.34	0.994	+1.08	−13.8

With no cosmological parameter left to adjust, SSEE fits the raw clustering nearly as well as the published model with three: three extra parameters buy $\Delta\chi^2 = +2.84$, less than the 3 they would return by chance, and $\Delta\mathrm{BIC} = -12.1$ in favour of SSEE. Between the two fixed backgrounds, at equal freedom, Planck's fits slightly better, $\chi^2_{\rm SSEE} - \chi^2_{\rm Planck} = +1.76$, which we state against our own interest; it comes entirely from $z = 0.38$ (+2.49), while at $z = 0.61$ SSEE fits better (−0.73). Each model carries its own neutrino mass ($\sum m_\nu = 0.06849\,\mathrm{eV}$ in SSEE, 0.06 eV in ΛCDM).

Outputs. The predicted $f\sigma_8$ of SSEE is 0.472 at $z = 0.38$ and 0.471 at $z = 0.61$ (0.477 and 0.470 for Planck's ΛCDM), with $\sigma_8 = 0.8153$. These are outputs of each model, reported to show what each sees, not the comparison.

Clustering does not set the amplitude. On its own, BOSS places the amplitude below the CMB value: the free ΛCDM control gives $\sigma_8 = 0.726 \pm 0.045$, and the σ_8 SSEE carries from its CMB lies $+1.99\sigma$ above it ($+1.90\sigma$ for Planck's). Yet imposing the CMB amplitude, with the whole background fixed, costs at most +2.84 in χ^2 against the free fit: the linear galaxy bias absorbs almost all of the difference. Galaxy clustering is therefore consistent with the CMB amplitude and does not vote on it.

6.1 Why the exploratory linear-Kaiser stage does not publish

An earlier, exploratory pass over the same data used linear Kaiser redshift-space distortions rather than the one-loop model reported above. Measured against the published BOSS DR12 consensus values (Alam et al. 2017 [Alam et al., 2017]) with the

growth rate left free, as in the official analysis, that pass reproduced the published values within 0.92σ on average (χ^2/dof= 1.048 at $k_{\rm max} = 0.08\,h$/Mpc; archived log `archive/logs_superados/boss_kaiser_20260808/`), a first check of the data reading, window convolution and Alcock–Paczynski deformation. It is quoted as history only: re-run, its minimiser is ill-conditioned along a flat nuisance direction and $\sigma(f\sigma_8)$ moves by 2–25%, so it is not a result of this paper. The linear model also under-predicts $f\sigma_8$ by 9–16% (increasing with z) once matched against small-scale modes ($k > 0.08\,h$/Mpc), a known limitation of first-order perturbation theory in the mildly non-linear regime. The production run for this paper uses a one-loop effective-theory model (velocileptors, Chen, Vlah & White 2020 [Chen et al., 2020]) that resolves this limitation, and §6 above reports its output, not the linear-theory exploratory result.

7 Conclusions

The growth sector of SSEE has a single matter density, $\Omega_m = 0.308881$, entering both the background and the growth of structure with no partition, no additional particle, and no non-minimal coupling parameter. Measured against raw survey data — rather than against ΛCDM-conditioned summary statistics — it has no S_8 tension. On KiDS-Legacy the amplitude the shear asks for lies -0.25σ from the one the model's own CMB fit supplies; with that amplitude fixed the growth sector has no free cosmological parameter, the fit costs almost nothing ($\chi^2_{\rm min} = 415.59$ against 414.99), and $S_8 = 0.8273$ is predicted against 0.8265 ± 0.0176 measured. (The earlier KiDS-1000 release, superseded, is kept as the record of how the retraction was reached.) Against raw BOSS DR12 clustering, once a free ΛCDM control reproduces the published analysis, SSEE with every cosmological parameter fixed fits the multipoles with $\Delta\chi^2 = +2.84$ against that model's three extra parameters (ΔBIC $= -12.1$), and clustering is consistent with the CMB amplitude without constraining it. The primordial amplitude is therefore counted once, at the CMB, and carried unchanged to every late-time probe. Cosmic shear and clustering do not discriminate between the backgrounds; what separates them is the parameter count.

The methodological point (§3) is offered as a general caution independent of SSEE: any model without background-level free parameters cannot exploit the mechanism by which combined datasets average away their own internal tensions, and is therefore a comparatively clean probe of whether those tensions are real.

References

S. Alam et al. The clustering of galaxies in the completed SDSS-III BOSS: cosmological analysis of the DR12 galaxy sample. *Mon. Not. Roy. Astron. Soc.*, 470:2617, 2017. doi: 10.1093/mnras/stx721.

C. Alcock and B. Paczynski. An evolution free test for non-zero cosmological constant. *Nature*, 281:358, 1979. doi: 10.1038/281358a0.

A. Amon et al. Dark energy survey year 3 results: cosmology from cosmic shear and robustness to modeling uncertainty. *Phys. Rev. D*, 105:023514, 2022. doi: 10.1103/PhysRevD.105.023514.

M. Asgari et al. KiDS-1000 cosmology: cosmic shear constraints and comparisons between two point statistics. *Astron. Astrophys.*, 645:A104, 2021. doi: 10.1051/0004-6361/202039070.

D. Baumann, A. Nicolis, L. Senatore, and M. Zaldarriaga. Cosmological non-linearities as an effective fluid. *J. Cosmol. Astropart. Phys.*, 2012(07):051, 2012. doi: 10.1088/1475-7516/2012/07/051.

F. Beutler and P. McDonald. Unified galaxy power spectrum measurements from 6dfgs, boss, and eboss. *JCAP*, 2021(11):031, 2021.

F. Beutler et al. The clustering of galaxies in the completed SDSS-III BOSS sample: anisotropic galaxy clustering in Fourier-space. *Mon. Not. Roy. Astron. Soc.*, 466:2242, 2017. doi: 10.1093/mnras/stw3298.

S. Bridle and L. King. Dark energy constraints from cosmic shear power spectra: impact of intrinsic alignments on photometric redshift requirements. *New J. Phys.*, 9:444, 2007. doi: 10.1088/1367-2630/9/12/444.

S.-F. Chen, Z. Vlah, and M. White. Consistent modeling of velocity statistics and redshift-space distortions in one-loop perturbation theory. *J. Cosmol. Astropart. Phys.*, 2020 (07):062, 2020. doi: 10.1088/1475-7516/2020/07/062.

S.-F. Chen, Z. Vlah, and M. White. A new analysis of galaxy 2-point functions in the BOSS survey, including full-shape information and post-reconstruction BAO. *J. Cosmol. Astropart. Phys.*, 2022(02):008, 2022. doi: 10.1088/1475-7516/2022/02/008.

M. Chevallier and D. Polarski. Accelerating universes with scaling dark matter. *Int. J. Mod. Phys. D*, 10:213, 2001. doi: 10.1142/S0218271801000822.

E. Di Valentino et al. Cosmology intertwined III: $f\sigma_8$ and s_8. *Astropart. Phys.*, 131:102604, 2021. doi: 10.1016/j.astropartphys.2021.102604.

F. Feroz, M. P. Hobson, and M. Bridges. MultiNest: an efficient and robust Bayesian inference tool for cosmology and particle physics. *Mon. Not. Roy. Astron. Soc.*, 398: 1601, 2009. doi: 10.1111/j.1365-2966.2009.14548.x.

C. Heymans et al. KiDS-1000 cosmology: multi-probe weak gravitational lensing and spectroscopic galaxy clustering constraints. *Astron. Astrophys.*, 646:A140, 2021. doi: 10.1051/0004-6361/202039063.

R. E. Kass and A. E. Raftery. Bayes factors. *J. Am. Statist. Assoc.*, 90:773, 1995. doi: 10.1080/01621459.1995.10476572.

E. V. Linder. Exploring the expansion history of the universe. *Phys. Rev. Lett.*, 90:091301, 2003. doi: 10.1103/PhysRevLett.90.091301.

A. J. Mead, J. A. Peacock, C. Heymans, S. Joudaki, and A. F. Heavens. An accurate halo model for fitting non-linear cosmological power spectra and baryonic feedback models. *Mon. Not. Roy. Astron. Soc.*, 454:1958, 2015. doi: 10.1093/mnras/stv2036.

Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astron. Astrophys.*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.

J. Torrado and A. Lewis. Cobaya: Code for Bayesian Analysis of hierarchical physical models, 2021.

A. H. Wright et al. KiDS-Legacy: Cosmological constraints from cosmic shear with the complete Kilo-Degree Survey. *Astron. Astrophys.*, 703:A158, 2025.